Clinical Infectious Diseases

IDSA GUIDELINES

2025 Clinical Practice Guideline Update by the Infectious Diseases Society of America on Histoplasmosis: Treatment of Mild or Moderate Acute Pulmonary Histoplasmosis in Adults, Children, and Pregnant People

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This article is part of a larger clinical practice guideline on the management of histoplasmosis in adults, children, and pregnant people, developed by the Infectious Diseases Society of America. In this article, the panel provides recommendations for treatment of mild and moderate acute pulmonary histoplasmosis. The panel's recommendations are based upon evidence derived from systematic literature reviews and adhere to a standardized methodology for rating the certainty of evidence and strength of recommendation according to the GRADE (Grading of Recommendations Assessment, Development, and Evaluation) approach.

Keywords. histoplasmosis; itraconazole; treatment; antifungals; guideline.

In patients presenting with mild or moderate acute pulmonary histoplasmosis, should antifungal treatment be given for resolution of symptoms?

Recommendation: In immunocompetent adults and children presenting with mild acute pulmonary histoplasmosis, the panel suggests against routinely providing antifungal treatment *(conditional recommendation, very low certainty of evidence)*.

Remarks:

- Treatment may be considered in immunocompetent patients with mild acute pulmonary histoplasmosis and prolonged duration of illness, progression of pulmonary infiltrates, or enlarging hilar or mediastinal adenopathy. In a large outbreak study, >75% of persons affected were ill for 1 week or less, and all recovered completely within 2 months without treatment [1].

Recommendation: In immunocompetent adults and children presenting with moderate acute pulmonary histoplasmosis, the panel suggests either antifungal treatment or no antifungal treatment, considering the severity and duration of signs/symptoms, as well as potential harms of antifungal treatment *(conditional recommendation, very low certainty of evidence)*.

Remarks:

- Moderate acute pulmonary histoplasmosis includes a heterogeneous group of patients. Prolonged duration of illness, worsening symptoms, progression of pulmonary infiltrates, enlarging hilar or mediastinal adenopathy, and more severe signs or symptoms favor treatment.
- Consider drug-drug interactions and other potential harms versus benefits of antifungal treatment when deciding whether to treat. Potential financial burden should be discussed with the patient as well.

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- The goals of treatment are to decrease the duration of illness and mitigate risk of dissemination, though treatment effectiveness in this patient population is unknown.
- When treatment is indicated, itraconazole is preferred [2].
- Initial dosing for original itraconazole capsules or oral solution: (adults: 200 mg three times daily for 3 days and then 200 mg twice daily for 6–12 weeks; children: 5 mg/kg/dose [up to a max of 200 mg/dose] three times daily for 3 days and then 5 mg/kg/dose twice daily [not to exceed 400 mg daily] for 6–12 weeks). Super-Bioavailable (SUBA) itraconazole (only available as capsules and currently approved for use in adults): 130 mg three times daily for 3 days, then 130 mg twice daily for 6–12 weeks. In consultation with a pharmacist, similar dosing for SUBA itraconazole based on the child's weight may be considered in children old enough to swallow capsules (as off-label use). For additional information on the various itraconazole formulations, see the Implementation Considerations section.
- Therapeutic drug monitoring (TDM) should be performed for patients receiving itraconazole [3–6]. In recent studies, approximately 20% of patients required dose adjustments due to sub- or super-therapeutic levels of itraconazole, and approximately 28% of patients experienced side effects [7, 8]. A goal trough concentration of itraconazole component >1 mg/L and <3–4 mg/L (as measured by chromatographic assay) is associated with efficacy and a lower risk of toxicity [3–7, 9–11]. Due to the long half-life of itraconazole, non-trough/random levels of itraconazole can also be used to monitor serum concentrations. Hydroxy-itraconazole is an active metabolite; however, a cutoff for combined hydroxy-itraconazole and itraconazole levels has not been established [10, 12, 13]. Patients with a combined hydroxy-itraconazole and itraconazole level >2 mg/L may respond similarly to patients with itraconazole levels >1 mg/L [14].
- Treatment of pregnant individuals should only be considered after carefully weighing the potential benefits versus harms of treatment, ideally in consultation with a maternal fetal medicine specialist and an infectious diseases specialist, as these cases are rare, complex, and highly variable. If treatment is necessary, azoles should be avoided in the first trimester when possible and liposomal amphotericin B used instead.

Recommendation: In immunocompromised adults and children presenting with mild or moderate acute pulmonary histoplasmosis who are at moderate to high risk of progression to disseminated disease, the panel suggests antifungal treatment *(conditional recommendation, very low certainty of evidence)*.

Remarks:

- Patients with asymptomatic or mild acute pulmonary histoplasmosis and a lesser degree of immunocompromise (see Table 1) may not warrant treatment.
- When treatment is indicated, itraconazole is preferred [2].
- Initial dosing for original itraconazole capsules or oral solution: (adults: 200 mg three times daily for 3 days and then 200 mg twice daily for 6–12 weeks; children: 5 mg/kg/dose [up to a max of 200 mg/dose] three times daily for 3 days and then 5 mg/kg/dose twice daily [not to exceed 400 mg daily] for 6–12 weeks). SUBA itraconazole (only available as capsules and currently approved for use in adults): 130 mg three times daily for 3 days, then 130 mg twice daily for 6–12 weeks. In consultation with a pharmacist, similar dosing for SUBA itraconazole based on the child´s weight may be considered in children old enough to swallow capsules (as off-label use). For additional information on the various itraconazole formulations, see the Implementation Considerations section.
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- Treatment of pregnant individuals should only be considered after carefully weighing the potential benefits versus harms of treatment, ideally in consultation with a maternal fetal medicine specialist and an infectious diseases specialist, as these cases are rare, complex, and highly variable. If treatment is necessary, azoles should be avoided in the first trimester when possible and liposomal amphotericin B used instead.

This article is part of a clinical practice guideline update on the treatment of pulmonary histoplasmosis in adults, children, and pregnant people, developed by the Infectious Diseases Society of America [34, 35]. These recommendations replace part of the previous recommendation on treatment of mild-to-moderate acute pulmonary histoplasmosis [2]. Alternative treatment options for patients who fail to improve, absorb, or are unable to tolerate first-line therapy will be addressed in a future, planned update. The primary audience for this recommendation is clinicians seeing patients with mild or moderate acute pulmonary histoplasmosis, including primary care clinicians, infectious diseases physicians, pulmonologists, specialists prescribing biologic response modifiers and other immunosuppressive agents, and cardiothoracic surgeons.

Table 1. Categories of Immunocompromise and Risk for Disseminated/Severe Histoplasmosis

Categories of immunocompromise represent a continuum rather than distinct categories. Conditions are categorized here as a guide; given limited evidence, this table is not exhaustive or exact.

High	Moderate	Low[a]
Receiving corticosteroids: [15] ≥2 mg/kg/d of prednisone (or equivalent) for persons ≤10 kg or ≥20 mg/d of prednisone (or equivalent) for persons >10 kg for at least 2 wks	Receiving corticosteroids: [15] 0.5–2 mg/kg/d of prednisone (or equivalent) for persons <10 kg or 5–20 mg/d of prednisone (or equivalent) for persons >10 kg for at least 4 wks	Receiving corticosteroids: [15] <0.5 mg/kg/d of prednisone (or equivalent) for persons <10 kg or ≤5 mg/d of prednisone (or equivalent) for persons >10 kg for at least 4 wks
Primary cellular immunodeficiency (eg, SCID, autosomal dominant hyperIgE syndrome [AD HIES], interferon-gamma receptor/IL-12 pathway defects)	Primary immunodeficiency (eg, common variable immunodeficiency, NF-kappaB pathway defects [NEMO], chronic mucocutaneous candidiasis, X-linked hyper IgM syndrome, autosomal recessive HIES)	
Advanced or untreated HIV/AIDS (CD4 <200 cells/mm^3)[b] [16]	HIV (CD4 200–300 cells/mm^3) [16–26]	HIV (CD4 ≥300 cells/mm^3); VL undetectable [16]
Hematopoietic stem cell transplant within 100 d or receiving immunosuppressive therapy for graft versus host disease	Hematopoietic stem cell transplant >100 d prior and no evidence of graft versus host disease	
	Hematologic malignancy	
CAR T-cell therapy within 90 d [27]	CAR T-cell therapy >90 d and resolved cytopenias [27]	
Solid organ transplant and treatment of rejection[c]	Solid organ transplant recipient on maintenance immunosuppressive regimen[c]	
Autoimmune and rheumatic diseases requiring treatment with biologic agents[d], especially those that interfere with T cell function and granuloma formation [23, 28–33]		Autoimmune and rheumatic diseases not requiring treatment
		General medical frailty, including but not limited to: Liver, kidney, lung disease, diabetes, malnutrition

Abbreviations: CAR, chimeric antigen receptor; HIV, human immunodeficiency virus; IL-12, interleukin 12; IgM, immunoglobulin M; VL, viral load.

[a]The following conditions confer no known increased risk: sickle cell disease and other asplenia syndromes; antibody, complement, or neutrophil deficiencies.

[b]Severe immunocompromise in children ≤5 y of age is defined as CD4+ T lymphocyte [CD4+] percentage <15%, and in individuals ≥6 y, CD4+ percentage <15% and CD4+ >200 lymphocytes/mm^3 [15].

[c]Carefully consider drug-drug interactions (eg, tacrolimus for Graft-vs-host disease [GVHD] prophylaxis).

[d]There are a variety of biologic agents with varying levels of immunosuppression. Serious infections have happened in patients receiving biologic response modifiers, including tuberculosis and disseminated infections caused by viruses, fungi, or bacteria. Frequently reported biologics associated with disseminated/severe histoplasmosis include: Tumor necrosis factor-alpha inhibitors (TNF-alpha inhibitors, eg, infliximab, etanercept, adalimumab); IL-12/IL-23 blockade (ustekinumab, risankizumab, guselkumab).

Mild acute pulmonary histoplasmosis refers to mild symptoms (eg, cough, fever, dyspnea, chest discomfort) that do not interfere with normal activities. Moderate acute pulmonary histoplasmosis refers to moderate symptoms significant enough to interfere with normal activities; patients with moderate disease may require low-flow oxygen supplementation and/or hospitalization.

METHODS

The panel's recommendations are based on evidence derived from systematic literature reviews and adhere to a standardized methodology for rating the certainty of evidence and strength of recommendation according to the GRADE (Grading of Recommendations Assessment, Development, and Evaluation) approach (Supplementary Figure 1) [36]. The recommendations have been endorsed by the Pediatric Infectious Diseases Society, the Society of Infectious Diseases Pharmacists, and the Mycoses Study Group Education and Research Consortium.

Strong recommendations, indicated by "the panel recommends," are made when the recommended course of action would apply to most people with few exceptions. Conditional recommendations, indicated by "the panel suggests," are made when the suggested course of action would apply to the majority of people with many exceptions and shared decision-making is important.

A comprehensive literature search (through January 2024) was conducted as part of a systematic review. Key eligibility criteria at both the topic and clinical question levels guided the search and selection of studies for inclusion (Supplementary Figure 2). For this question, a larger search on treatment of histoplasmosis was conducted. A critical appraisal of the evidence according to the GRADE approach, along with an assessment of the benefits and harms of care options informed the recommendations [36, 37]. Details of the systematic review and guideline development processes are available in the Supplementary Material.

SUMMARY OF EVIDENCE

Limited evidence was identified for the outcomes of mortality (9 studies [1, 38–45]), symptom resolution/radiographic

regression (9 studies [1, 38–41, 43–46]), and toxicity (1 study [39]) (Supplementary Tables 1 and 2). For the outcome of mortality, there were no deaths attributable to histoplasmosis in any study regardless of whether patients were treated. Five of 9 studies were retrospective reviews (including 2 school outbreak reports of 876 affected people, 708 of whom were symptomatic), 3 were case reports, and 1 was a prospective, nonrandomized trial. Many of the same studies reported on symptomatic resolution and/or radiographic regression. In one of the outbreak studies, 353 people (mostly adolescents) were symptomatic but received no treatment, and >75% were ill for 1 week or less, with all recovering within 2 months [1]. In another outbreak study, 523/682 study participants had serologic evidence of infection, 355 of whom developed symptoms, with only 13 receiving an antifungal agent [38]. In the largest outbreak, of over 100 000 presumed infected, 435 presented to a hospital (285 with acute respiratory histoplasmosis), with only 43 receiving treatment [47]. In this study, 7 immunocompromised individuals with disseminated histoplasmosis recovered without treatment. Only 1 study addressed toxicity within the context of this clinical question [39]. In this study, 37 patients with confirmed histoplasmosis were treated with itraconazole 200–400 mg/day for a median 9 months, and itraconazole was stopped in 1/37 patients due to toxicity.

In one of the outbreak cohorts [38], 4 people were receiving immunocompromising medications, and 3 of the 4 were hospitalized. A case report of a pediatric oncology patient with acute pulmonary histoplasmosis who received treatment with itraconazole reported improvement [40]. A retrospective review of 52 children with histoplasmosis (7 immunocompromised) reported longer duration of antigenuria, antigenemia, and duration of therapy in immunocompromised patients, with no recurrence within 2 years for those treated with antifungals [43].

For all recommendations, the certainty of evidence is very low according to the GRADE approach due to study risk of bias (Supplementary Table 3), inconsistency amongst studies (eg, patients in an outbreak study have a higher inoculum, heterogeneous populations in terms of histoplasmosis severity and type), and imprecision (small number of events) [48–50]. Refer to the Supplementary Material for more information on each study and exact judgments affecting the certainty of evidence for each outcome.

RATIONALE FOR RECOMMENDATIONS

The studies demonstrate that most cases of mild-to-moderate acute pulmonary histoplasmosis will resolve without treatment. Data on the prevalence of *Histoplasma capsulatum* exposure, rate at which significant illness develops, and prognosis of nonsevere acute pulmonary disease largely derive from population-level histoplasmin sensitivity studies and large outbreak reports, most dating from the 1950s–80s. These indicate that exposure to *H. capsulatum* is widespread in endemic areas, with over half of children by age 8 [51] and more than 80% of long-term residing adults [52] demonstrating an immunologic response to *Histoplasma* antigen. Most infections are subclinical with an estimated fewer than 5% of exposed individuals developing even mild symptoms [46]. In a large urban outbreak study involving an estimated 120 000 individuals, only 435 cases (0.36%) were identified in a hospital setting (and therefore likely to meet current criteria for moderate or severe disease) [47]. At most, 4 of 285 individuals with acute respiratory disease required treatment, with all others recovering without treatment. In another large outbreak involving 383 junior high school students and some adult staff, only 1 patient required hospitalization, and all recovered without treatment, most within 2 weeks of symptom onset [1].

Treatment decisions also require an assessment of the potential harms of treatment. Itraconazole is associated with several common undesirable effects, including nausea and vomiting, rash, and peripheral edema. More serious complications, including hepatotoxicity and heart failure, are rare but have been reported. Itraconazole is a strong CYP3A4 inhibitor with many significant drug-drug interactions and is known to have highly variable oral absorption requiring monitoring of drug levels. Its use is contraindicated in the first trimester of pregnancy, necessitating involvement of maternal fetal medicine and infectious diseases specialists and use of alternative agents such as amphotericin B with potentially greater undesirable effects. Treatment duration of 6–12 weeks also imposes a large pill burden, most notably for children. Overall, the panel assesses the burden of undesirable effects to be typically small in children, moderate in adults, and large in pregnant people. Treatment courses may also be associated with significant cost with variable insurance coverage.

Since most immunocompetent patients with mild to moderate acute pulmonary histoplasmosis recover without specific treatment, and several potential harms of treatment are apparent, the panel agrees that the overall balance of benefits versus harms favors not treating with an antifungal medication in most cases. Greater consideration to treatment may be appropriate in cases with more prolonged symptom duration (eg, >1 month), progressive symptoms or radiographic abnormalities, or more severe initial symptoms. Treatment in these scenarios would be intended to shorten duration of symptoms, prevent progression to more severe disease, and/or prevent late sequalae of pulmonary histoplasmosis such as mediastinal granuloma or fibrosing mediastinitis. However, there are no data demonstrating treatment is associated with any of these potentially improved clinical outcomes.

Progression to severe acute pulmonary histoplasmosis or disseminated disease appears rare in immunocompetent patients in large outbreak studies. However, immunocompromise has been identified as a clear risk factor for more severe disease

in these and other reports [53]. For this reason, the panel agrees that the overall balance of benefits to harms favors antifungal treatment in patients with acute pulmonary histoplasmosis and an underlying immunocompromise that places them at significant risk for disease progression.

IMPLEMENTATION CONSIDERATIONS

Proper implementation of the approaches suggested above has two major limitations [54]. First, acute mild to moderate histoplasmosis is vastly underdiagnosed, including cases that occur within immunocompromised patients, because of the non-specificity of symptoms and the high rate of spontaneous resolution. As such, most cases of acute histoplasmosis remain undetected. The second limitation to implementation is related to the first, that is, lack of recognition. The remedy to this involves creating greater awareness of histoplasmosis as a frequent cause of community-acquired pneumonia, especially in highly endemic regions. Current efforts by the Centers for Disease Control and Prevention (eg, Fungal Disease Awareness Week) and other groups such as the Mycoses Study Group, aim to increase awareness of histoplasmosis and other endemic mycoses through programs which encourage a lower threshold for rapid testing (eg, urine *Histoplasma* antigen) and an overall greater awareness of this pathogen as a cause of non-specific community-acquired pneumonia. In the absence of specific laboratory testing, greater emphasis must be placed on patient history, including travel, occupation, hobbies, and detailed history of any significant comorbidities which might increase the risk of more aggressive disease.

There are currently three available oral formulations of itraconazole that are not interchangeable, as there are differences in dosing, administration, and cost. The original capsule formulation has limited and variable oral bioavailability, and it must be taken with food containing high fat content [55]. Its bioavailability may be increased with a low stomach pH and longer gastric retention time. Although slightly more costly than the capsule formulation, the oral solution of itraconazole has better bioavailability [10, 56]. However, this formulation must be taken on an empty stomach and is associated with gastrointestinal side effects such as osmotic diarrhea due to the presence of hydroxypropyl-β-cyclodextrin, used to solubilize the drug. A newer oral capsule formulation of itraconazole, SUBA itraconazole, is now available. Currently the most expensive of the 3, this formulation contains itraconazole in a polymeric matrix that enhances its bioavailability compared the original capsule formulation and lessens the impact of gastric pH (and pH-altering medications) on absorption [57–59]. Similar to the original capsule formulation, it is recommended to take SUBA itraconazole with food.

Therapeutic drug monitoring should be performed for patients receiving itraconazole [3–6]. In general, blood concentrations should be checked after approximately 5–7 days when loading doses are used, and 10–14 days without a loading dose [6]. Levels should also be checked when interacting drugs are started or stopped, there are concerns for patient adherence or gastrointestinal absorption, and/or the patient has symptoms of toxicities. A goal trough concentration of itraconazole component >1 mg/L and <3–4 mg/L (as measured by chromatographic assay) is associated with efficacy and a lower risk of toxicity [3–6, 9–11]. Due to the long half-life of itraconazole, non-trough/random levels of itraconazole can also be used to monitor serum concentrations. Hydroxy-itraconazole is an active metabolite; however, a cutoff for combined hydroxy-itraconazole and itraconazole levels has not been established [10, 12, 13]. Patients with a combined hydroxy-itraconazole and itraconazole level >2 mg/L may respond similarly to patients with itraconazole levels >1 mg/L [14].

RESEARCH NEEDS

Randomized controlled trials or large cohort studies evaluating outcomes of antifungal treatment (including itraconazole and alternative azoles) versus no antifungal treatment in patients with mild-to-moderate acute pulmonary histoplasmosis would be very helpful. However, studies of this sort are extraordinarily difficult to conduct, largely due to the difficulty in diagnosing acute disease and identifying subgroups who might benefit from therapeutic intervention. Clinical trials of this sort involving older, generic antifungals are likely to require sponsorship and/or coordination by a federal agency, such as the Centers for Disease Control and Prevention or the National Institute of Allergy and Infectious Diseases, on the basis of public health needs. At present, there are no randomized controlled trials involving treatment of mild or moderate pulmonary histoplasmosis. As a result, the impact of treatment on symptom duration, progression to severe pulmonary histoplasmosis or disseminated disease, or development of late sequelae of pulmonary histoplasmosis is largely unknown, and the above recommendations are largely based on expert interpretation of observational studies.

Supplementary Data

Supplementary materials are available at *Clinical Infectious Diseases* online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

Notes

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Sandra Arnold and Andrej Spec are chair and vice chair, respectively, of the expert panel. Robert Lentz and Peter Pappas served as clinical leads for the question addressed in this manuscript. Kayla Stover and Nathan Wiederhold led the development of remarks on therapeutic drug monitoring for itraconazole. Remaining panelists assisted with conception and design of the analysis, interpretation of data, drafting and revising the recommendations and manuscript, and final approval of the recommendations and manuscript to be published. Jennifer Loveless, methodologist, was responsible for general project management, synthesizing and presenting the data, and leading the panel according to the GRADE process.

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Additional information. More detailed information on the analysis and development of recommendations is available in the Supplementary Material.

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Potential conflicts of interest. Evaluation of relationships as potential conflicts of interest (COIs) is determined by a review process. The assessment of disclosed relationships for possible COIs is based on the relative weight of the financial relationship (ie, monetary amount) and the relevance of the relationship (ie, the degree to which an association might reasonably be interpreted by an independent observer as related to the topic or recommendation of consideration). The following panelists have scientific advisory/consultant roles related to the topic (ie, manageable conflicts of interest): A. S. (vice chair) with Mayne and Scynexis, C. A. K. with Cidara Therapeutics (concluded) and Laboratoires SMB (concluded), K. S. with Cidara Therapeutics (concluded), and N. W. with F2G (concluded). C. A. K. is Editor-in-Chief for UpToDate's Infectious Diseases sections, and J. B. is an editor for UpToDate's Fungal Infections sections. The following panelists have research relationships related to the topic: A. S. (vice chair) with Astellas; R. M. with Scynexis and F2G; P. P. with Mayne; and N. W. with F2G, Scynexis, Mycovia (concluded), and Sfunga (concluded). No disclosures were reported for all other authors (the majority of panelists) including chair. All other authors report no potential conflicts.

All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

References

1. Brodsky AL, Gregg MB, Loewenstein MS, Kaufman L, Mallison GF. Outbreak of histoplasmosis associated with the 1970 Earth Day activities. Am J Med **1973**; 54: 333–42.
2. Wheat LJ, Freifeld AG, Kleiman MB, et al. Clinical practice guidelines for the management of patients with histoplasmosis: 2007 update by the Infectious Diseases Society of America. Clin Infect Dis **2006**; 45:807–25.
3. Ashbee HR, Barnes RA, Johnson EM, Richardson MD, Gorton R, Hope WW. Therapeutic drug monitoring (TDM) of antifungal agents: guidelines from the British Society for Medical Mycology. J Antimicrob Chemother **2014**; 69: 1162–76.
4. Chau MM, Daveson K, Alffenaar J-WC, et al. Consensus guidelines for optimising antifungal drug delivery and monitoring to avoid toxicity and improve outcomes in patients with haematological malignancy and haemopoietic stem cell transplant recipients, 2021. Intern Med J **2021**; 51 Suppl 7:37–66.
5. Laverdiere M, Bow EJ, Rotstein C, et al. Therapeutic drug monitoring for triazoles: a needs assessment review and recommendations from a Canadian perspective. Can J Infect Dis Med Microbiol **2014**; 25:327–43.
6. McCreary EK, Davis MR, Narayanan N, et al. Utility of triazole antifungal therapeutic drug monitoring: insights from the Society of Infectious Diseases Pharmacists: endorsed by the Mycoses Study Group Education and Research Consortium. Pharmacotherapy **2023**; 43:1043–50.
7. Osborn MR, Zuniga-Moya JC, Mazi PB, Rauseo AM, Spec A. Side effects associated with itraconazole therapy. J Antimicrob Chemother **2025**; 80:503–8.
8. Spec A, Thompson GR, Miceli MH, et al. MSG-15: super-bioavailability itraconazole versus conventional itraconazole in the treatment of endemic mycoses—a multicenter, open-label, randomized comparative trial. Open Forum Infect Dis **2024**; 11:ofae010.
9. Andes D, Pascual A, Marchetti O. Antifungal therapeutic drug monitoring: established and emerging indications. Antimicrob Agents Chemother **2009**; 53:24–34.
10. Cartledge JD, Midgely J, Gazzard BG. Itraconazole solution: higher serum drug concentrations and better clinical response rates than the capsule formulation in acquired immunodeficiency syndrome patients with candidosis. J Clin Pathol **1997**; 50:477–80.
11. Lestner JM, Roberts SA, Moore CB, Howard SJ, Denning DW, Hope WW. Toxicodynamics of itraconazole: implications for therapeutic drug monitoring. Clin Infect Dis **2009**; 49:928–30.
12. Firkus D, Abu Saleh OM, Enzler MJ, et al. Does metabolite matter? Defining target itraconazole and hydoxy-itraconazole serum concentrations for blastomycosis. Mycoses **2023**; 66:412–9.
13. Khurana A, Agarwal A, Singh A, et al. Predicting a therapeutic cut-off serum level of itraconazole in recalcitrant tinea corporis and cruris—a prospective trial. Mycoses **2021**; 64:1480–8.
14. Wiederhold NP, Schwartz IS, Patterson TF, Thompson GR III. Variability of hydroxy-itraconazole in relation to itraconazole bloodstream concentrations. Antimicrob Agents Chemother **2021**; 65:e02353-20.
15. Kroger A, Bahta L, Long S, et al. General best practice guidelines for immunization: altered immunocompetence. Available at: https://www.cdc.gov/vaccines/hcp/imz-best-practices/alteredimmunocompetence.html. Accessed 16 June 2024..
16. Panel on Guidelines for the Prevention and Treatment of Opportunistic Infections in Adults and Adolescents with HIV. Guidelines for the prevention and treatment of opportunistic infections in adults and adolescents with HIV. Available at: https://clinicalinfo.hiv.gov/en/guidelines/adult-and-adolescent-opportunistic-infection/whats-new-guidelines. Accessed 16 June 2024.
17. Ashbee HR, Evans EGV, Viviani MA, et al. Histoplasmosis in Europe: report on an epidemiological survey from the European Confederation of Medical Mycology Working Group. Med Mycol **2008**; 46:57–65.
18. Antinori S, Magni C, Nebuloni M, et al. Histoplasmosis among human immunodeficiency virus-infected people in Europe: report of 4 cases and review of the literature. Medicine **2006**; 85:22–36.
19. Anderson AM, Mehta AK, Wang YF, et al. HIV-associated histoplasmosis in a nonendemic area of the United States during the HAART era: role of migration from endemic areas and lack of antiretroviral therapy. J Int Assoc Physicians AIDS Care **2010**; 9:296–300.

20. Bourgeois N, Douard-Enault C, Reynes J, et al. Seven imported histoplasmosis cases due to *Histoplasma capsulatum* var. capsulatum: from few weeks to more than three decades asymptomatic period. J Mycol Med **2011**; 21:19–23.
21. Buitrago MJ, Bernal-Martinez L, Castelli MV, et al. Histoplasmosis and paracoccidioidomycosis in a non-endemic area: a review of cases and diagnosis. J Travel Med **2011**; 18:26–33.
22. Choi J, Nikoomanesh K, Uppal J, Wang S. Progressive disseminated histoplasmosis with concomitant disseminated nontuberculous mycobacterial infection in a patient with AIDS from a nonendemic region (California). BMC Pulm Med **2019**; 19:46.
23. Gandhi V, Ulyanovskiy P, Epelbaum O. Update on the spectrum of histoplasmosis among Hispanic patients presenting to a New York City municipal hospital: a contemporary case series. Respir Med Case Rep **2015**; 16:60–4.
24. Peigne V, Dromer F, Elie C, Lidove O, Lortholary O; French Mycosis Study Group. Imported acquired immunodeficiency syndrome-related histoplasmosis in metropolitan France: a comparison of pre-highly active anti-retroviral therapy and highly active anti-retroviral therapy eras. Am J Trop Med Hyg **2011**; 85: 934–41.
25. Martin-Iguacel R, Kurtzhals J, Jouvion G, Nielsen SD, Llibre JM. Progressive disseminated histoplasmosis in the HIV population in Europe in the HAART era. Case report and literature review. Infection **2014**; 42:611–20.
26. Norman FF, Martin-Davila P, Fortun J, et al. Imported histoplasmosis: two distinct profiles in travelers and immigrants. J Travel Med **2009**; 16:258–62.
27. Shahid Z, Jain T, Dioverti V, et al. Best practice considerations by the American Society of Transplant and Cellular Therapy: infection prevention and management after chimeric antigen receptor T cell therapy for hematological malignancies. Transplant Cell Ther **2024**; 30:955–69.
28. Hage CA, Bowyer S, Tarvin SE, Helper D, Kleiman MB, Wheat LJ. Recognition, diagnosis, and treatment of histoplasmosis complicating tumor necrosis factor blocker therapy. Clin Infect Dis **2010**; 50:85–92.
29. Jain VV, Evans T, Peterson MW. Reactivation histoplasmosis after treatment with anti-tumor necrosis factor alpha in a patient from a nonendemic area. Respir Med **2006**; 100:1291–3.
30. Lucey O, Carroll I, Bjorn T, Millar M. Reactivation of latent *Histoplasma* and disseminated cytomegalovirus in a returning traveller with ulcerative colitis. JMM Case Rep **2018**; 5:e005170.
31. Prakash K, Richman D. A case report of disseminated histoplasmosis and concurrent cryptococcal meningitis in a patient treated with ruxolitinib. BMC Infect Dis **2019**; 19:287.
32. Sani S, Bilal J, Varma E, Kollampare S, Sudano D. Not your typical Arizona granuloma: a case report of disseminated histoplasmosis. Am J Med **2018**; 131:e375–6.
33. Wallis RS, Broder MS, Wong JY, Hanson ME, Beenhouwer DO. Granulomatous infectious diseases associated with tumor necrosis factor antagonists. Clin Infect Dis **2004**; 38:1261–5.
34. Arnold S, Spec A, Baddley JW, et al. 2025 clinical practice guideline update by the Infectious Diseases Society of America on histoplasmosis: treatment of asymptomatic *Histoplasma* pulmonary nodules (histoplasmomas) and mild or moderate acute pulmonary histoplasmosis in adults, children, and pregnant people. Clin Infect Dis **2025**.
35. Baddley JW, Wolf J, Ardura MI, et al. 2025 clinical practice guideline update by the Infectious Diseases Society of America on histoplasmosis: treatment of asymptomatic *Histoplasma* pulmonary nodules (histoplasmomas) in adults, children, and pregnant people. Clin Infect Dis **2025**.
36. Guyatt GH, Oxman AD, Vist GE, et al. GRADE Working Group. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ **2008**; 336:924–6.
37. Infectious Diseases Society of America. IDSA Handbook on Clinical Practice Guideline Development. Available at: https://www.idsociety.org/practice-guideline/clinical-practice-guidelines-development-training-and-resources/. Accessed 03 January 2024.
38. Chamany S, Mirza SA, Fleming JW, et al. A large histoplasmosis outbreak among high school students in Indiana, 2001. Pediatr Infect Dis J **2004**; 23:909–14.
39. Dismukes WE, Bradsher RW, Cloud GC, et al. Itraconazole therapy for blastomycosis and histoplasmosis. NIAID mycoses study group. Am J Med **1992**; 93: 489–97.
40. Hess J, Fondell A, Fustino N, et al. Presentation and treatment of histoplasmosis in pediatric oncology patients: case series and review of the literature. J Pediatr Hematol Oncol **2017**; 39:137–40.
41. Muhi S, Crowe A, Daffy J. Acute pulmonary histoplasmosis outbreak in a documentary film crew travelling from Guatemala to Australia. Trop Med Infect Dis **2019**; 4:25.
42. Olson TC, Bongartz T, Crowson CS, Roberts GD, Orenstein R, Matteson EL. Histoplasmosis infection in patients with rheumatoid arthritis, 1998–2009. BMC Infect Dis **2011**; 11:145.
43. Ouellette CP, Stanek JR, Leber A, Ardura MI. Pediatric histoplasmosis in an area of endemicity: a contemporary analysis. J Pediatric Infect Dis Soc **2019**; 8:400–7.
44. Staffolani S, Riccardi N, Farina C, et al. Acute histoplasmosis in travelers: a retrospective study in an Italian referral center for tropical disease. Pathog Glob Health **2020**; 114:40–5.
45. Agossou M, Turmel J-M, Aline-Fardin A, Venissac N, Desbois-Nogard N. Acute pulmonary histoplasmosis of immunocompetent subjects from Martinique, Guadeloupe and French Guiana: a case series. BMC Pulm Med **2023**; 23:95.
46. Wheat LJ. Diagnosis and management of histoplasmosis. Eur J Clin Microbiol Infect Dis **1989**; 8:480–90.
47. Wheat LJ, Slama TG, Eitzen HE, Kohler RB, French ML, Biesecker JL. A large urban outbreak of histoplasmosis: clinical features. Ann Intern Med **1981**; 94:331–7.
48. Sterne JA, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ **2016**; 355:i4919.
49. Moola S, Munn Z, Tufanaru C, et al. Systematic reviews of etiology and risk (2020). In: Aromataris E, Lockwood C, Porritt K, et al. eds. JBI manual for evidence synthesis. JBI, 2024. doi:10.46658/JBIMES-24-06.
50. McGuinness LA, Higgins JPT. Risk-of-bias VISualization (robvis): an R package and shiny web app for visualizing risk-of-bias assessments. Res Synth Methods **2021**; 12:55–61.
51. Christie A, Peterson JC. Histoplasmin sensitivity. J Pediatr **1946**; 29:417–32.
52. Edwards LB, Acquaviva FA, Livesay VT. Further observations on histoplasmin sensitivity in the United States. Am J Epidemiol **1973**; 98:315–25.
53. Sathapatayavongs B, Batteiger BE, Wheat J, Slama TG, Wass JL. Clinical and laboratory features of disseminated histoplasmosis during two large urban outbreaks. Medicine (Baltimore) **1983**; 62:263–70.
54. Smith DJ, Free RJ, Thompson GR III, et al. Clinical testing guidance for coccidioidomycosis, histoplasmosis, and blastomycosis in patients with community-acquired pneumonia for primary and urgent care providers. Clin Infect Dis **2024**; 78:1559–63.
55. Zimmermann T, Yeates RA, Laufen H, Pfaff G, Wildfeuer A. Influence of concomitant food intake on the oral absorption of two triazole antifungal agents, itraconazole and fluconazole. Eur J Clin Pharmacol **1994**; 46:147–50.
56. Koks CHW, Meenhorst PL, Bult A, Beijnen JH. Itraconazole solution: summary of pharmacokinetic features and review of activity in the treatment of fluconazole-resistant oral candidosis in HIV-infected persons. Pharmacol Res **2002**; 46:195–201.
57. Abuhelwa AY, Foster DJR, Mudge S, Hayes D, Upton RN. Population pharmacokinetic modeling of itraconazole and hydroxyitraconazole for oral SUBA-itraconazole and sporanox capsule formulations in healthy subjects in fed and fasted states. Antimicrob Agents Chemother **2015**; 59:5681–96.
58. Thompson GR III, Lewis P, Mudge S, Patterson TF, Burnett BP. Open-label crossover oral bioequivalence pharmacokinetics comparison for a 3-day loading dose regimen and 15-day steady-state administration of SUBA-itraconazole and conventional itraconazole capsules in healthy adults. Antimicrob Agents Chemother **2020**; 64:e00400–20.
59. Nield B, Larsen SR, van Hal SJ. Clinical experience with new formulation SUBA®-itraconazole for prophylaxis in patients undergoing stem cell transplantation or treatment for haematological malignancies. J Antimicrob Chemother **2019**; 74:3049–55.